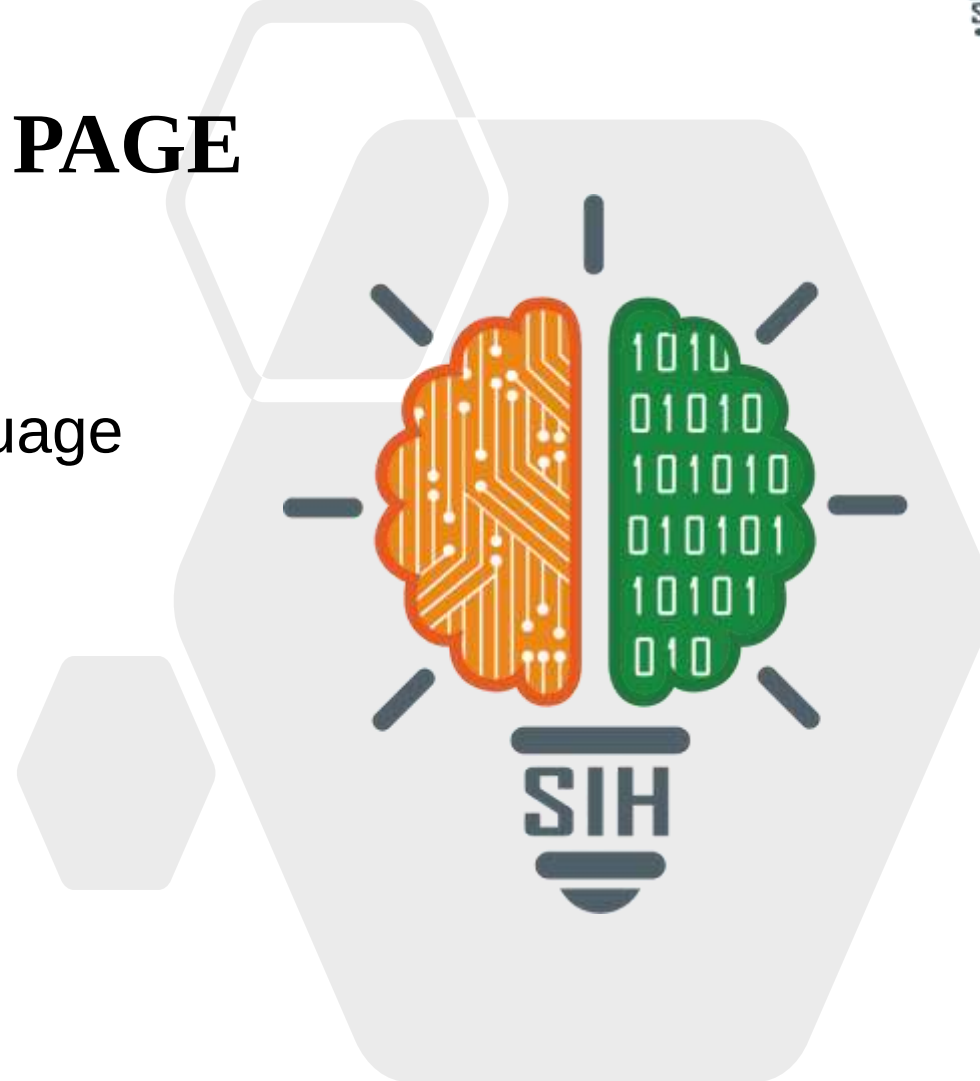


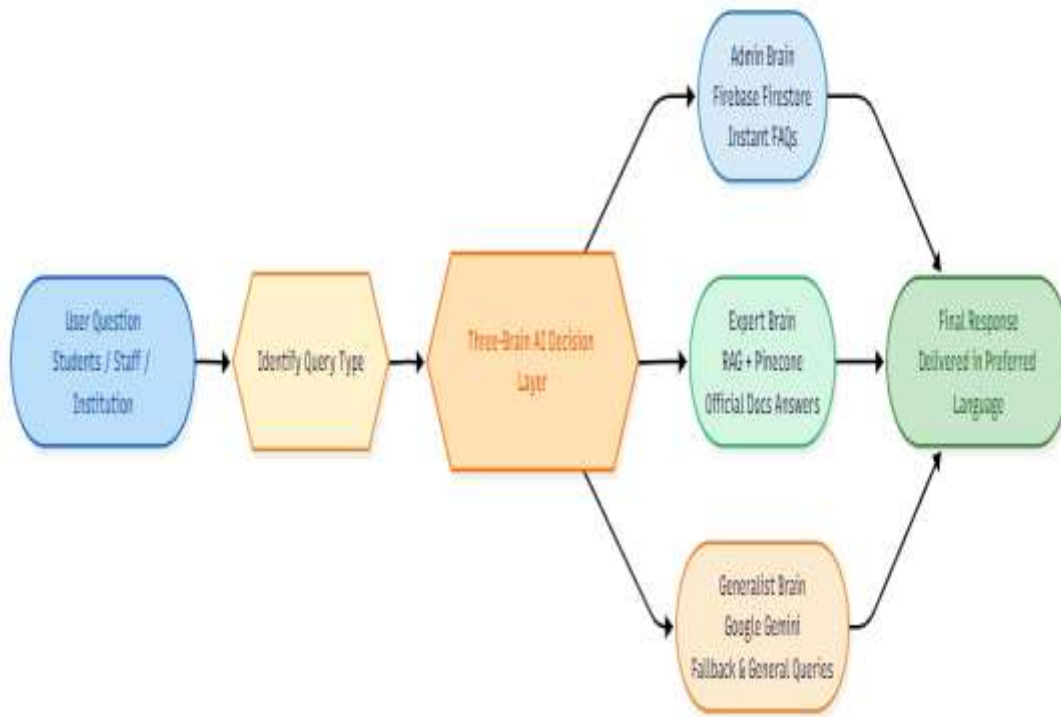
TITLE PAGE

- **Problem Statement ID** – 25104
- **Problem Statement Title-** Language
Agnostic Chatbot
- **Theme-** Smart Education
- **PS Category-** Software
- **Team ID-** 73471
- **Team Name-** Trojan_Source#



IDEA TITLE

Proposed Solution



Step 1: User Interaction

Students, faculty, or staff raise a query through the chatbot interface in their preferred language.

Step 2: Problem Context

For Students: Provides 24/7 academic and administrative support.
For Staff: Reduces repetitive workload by handling routine queries.
For Institutions: Enhances communication, improves efficiency, and builds a digital-first campus environment.

Step 3: Decision Layer – Three-Brain AI Architecture

Admin Brain (Firestore Firestore): Handles FAQs and routine administrative queries with instant, accurate responses.

Expert Brain (RAG Pipeline + Pinecone): Retrieves information directly from official documents (policies, circulars, PDFs, PPTs) to ensure factual and verified answers.

Generalist Brain (Google Gemini): Acts as a fallback for general or off-topic questions, enabling natural, conversational interactions.

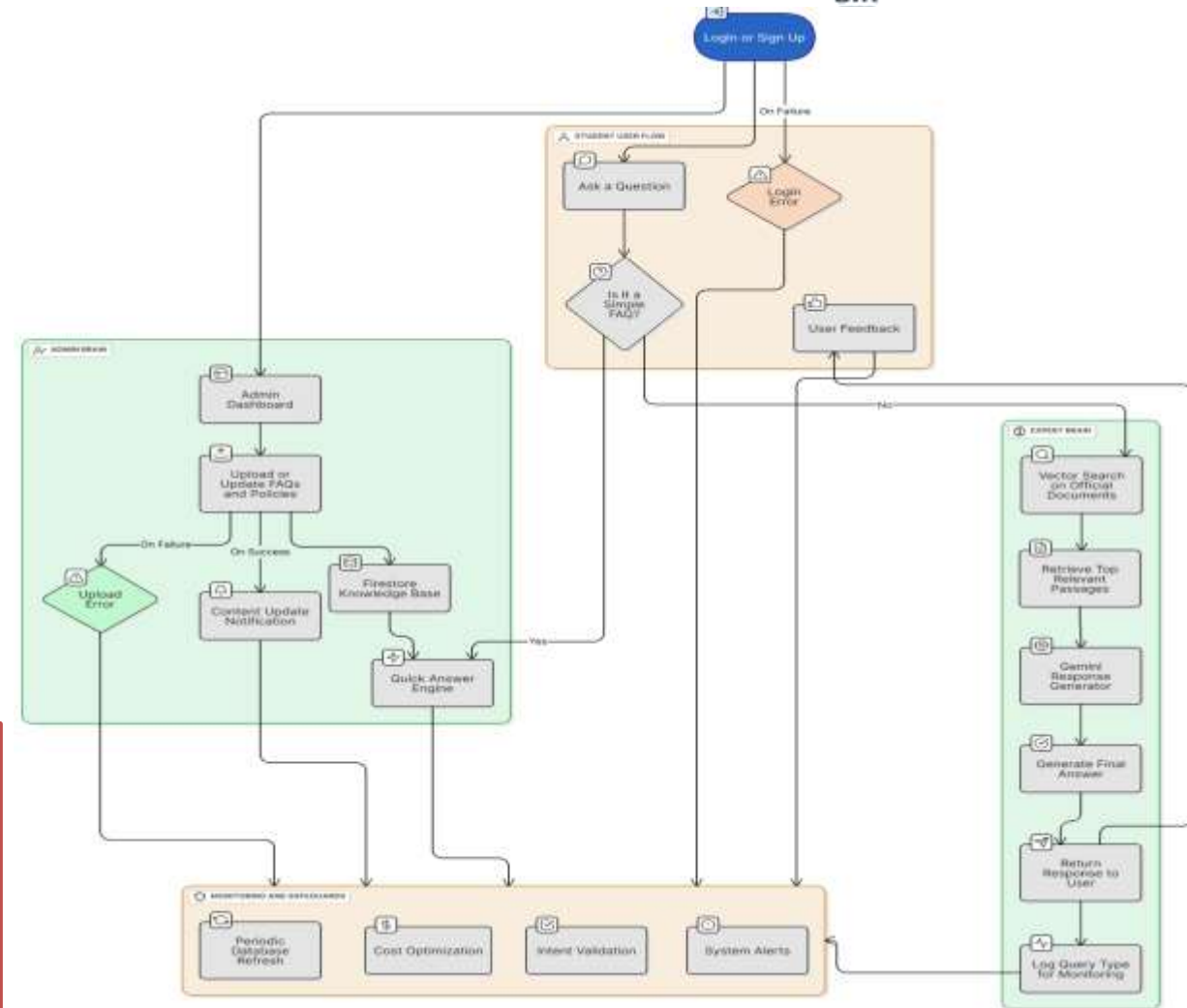
Step 4: Response Delivery

The selected brain generates a response which is translated (if required) and delivered back to the user in their preferred language.

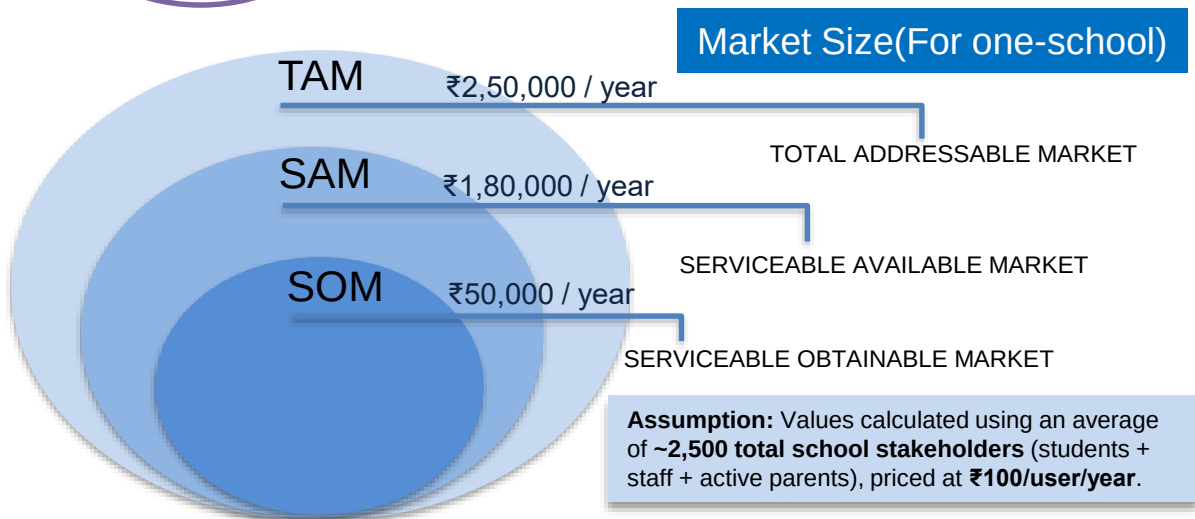
Our Workflow

- ❖ **Intent:** Dialogflow understands the user's query.
- ❖ **Routing:** Simple FAQs go to the **Admin Brain**; complex queries go to the **Expert Brain**.
- ❖ **Answer:** The correct brain finds or generates the answer.
- ❖ **Response:** The final answer is delivered to the user.

Tech Stack



FEASIBILITY AND VIABILITY



Cost Structure & Revenue Streams

- **Technology & Development Costs:** Development of chatbot platform, integration with Firestore, RAG pipeline, and Gemini API usage.
- **Maintenance Costs:** Regular updates to database (FAQs + docs), cloud hosting charges, and API monitoring.
- **Operational Costs:** Training data updates, server scaling, and admin dashboard support.
- **Revenue Streams:**
 - Subscription fees from institutions for chatbot deployment.
 - Premium features like multilingual support, analytics dashboard, and ERP integration.
 - Consultancy charges for customization (custom workflows, integrations).

Challenges

Data Accuracy & Freshness

NLU Intent Conflict at Scale

Managing Generative AI Costs

Preventing Hallucinations

Solutions

“Two-Brain” approach with **Admin Brain (Firestore dashboard)** for easy updates + **Expert Brain (RAG)** for official docs refresh.

Use **Dialogflow’s Validation Tool** + 10–15 unique training phrases per intent to avoid overlap.

Cost-efficient routing — common queries go to **Admin Brain**; only complex queries use **Gemini Expert Brain**.

Strict **contextual grounding** in prompts — answers must come **only from vector DB sources**.

IMPACT & BENEFITS



Student

Get **24/7 instant answers** to queries in multiple languages, with voice and OCR support. Experience **reduced wait times** and lower academic stress.

Teachers/Faculty

Repetitive queries are **automated**, saving valuable teaching time. Gain **data-driven insights** to better support students.

DTE (Directorate of Technical Education)

Enables **modern, standardized communication** across colleges. Creates a **single platform for unified support and monitoring**. Positions DTE as a **technology leader in digital education services**.

Social

Breaks language barriers, ensuring accessibility for diverse learners. Promotes **inclusivity and equal opportunities** for students from all backgrounds. Encourages **digital literacy and collaboration** within academic communities.

Economic

Automates **thousands of hours of manual work**, improving efficiency. Delivers **significant operational cost savings** to institutions. Contributes to **higher student retention and success rates**.

Environmental

Reduces paper waste by shifting notices and processes online. Encourages **digital-first practices**, cutting down on printing costs. Supports a **greener, more sustainable campus ecosystem**.

❖ Core AI Models:

- Google Gemini API: https://ai.google.dev/docs/gemini_api_overview
- Google Dialogflow ES: <https://cloud.google.com/dialogflow/es/docs>
- Google Cloud Vision API (OCR): <https://cloud.google.com/vision/docs/ocr>

❖ Key Architectural Pattern:

- Retrieval-Augmented Generation (RAG):
<https://www.google.com/search?q=https://research.facebook.com/blog/2020/09/retrieval-augmented-generation-for-knowledge-intensive-nlp-tasks/>

❖ Database Technologies:

- Firebase Firestore: <https://firebase.google.com/docs/firestore>
- Pinecone Vector DB: <https://www.pinecone.io/>

❖ Backend Framework:

- Flask (Python): <https://flask.palletsprojects.com/>